

Overview and Objectives

Concepts of the architecture of the web, websites, web applications, and driving these from information stored in databases. Topics include the structure of the web; simple web pages; using javascript to dynamically update web pages; using asynchronous background requests to a server to update pages; writing code on the server; creating, maintaining, and querying databases.

By the end of the course students will be able to

- Understand the basic infrastructure of the web and web servers.
- Create static web sites with html and css.
- Write and use CSS to properly layout the contents of a web page.
- Write client-side javascript to create dynamic web pages including single-page applications.
- Write client-side javascript with asynchronous functions (AJAX) to request background information from publicly available APIs
- Write server-side javascript to dynamically serve information in response to AJAX requests.
- Create databases with tables in a modern database system such as MySQL
- Write SQL queries to retrieve information from existing database tables.
- Write server-side javascript to interact with a database system
- Be able to use basic command-line interface commands to execute programs on a remote server.
- Be able to upload and modify files on a remote server.
- Create a full stack web application by dynamically building pages based on requests for information from a server which pulls data from a database.

Essential Information

Instructor: Erik Steinmetz
E-mail: steinmee@augsborg.edu
Phone: (612) 330-1062
Office: Hagfors 369E
Office Hours: MW 1:30 pm - 2:00 pm, T 10:30 am - 11:00 am, or by appointment
Class Location: Hagfors 367
Class Times: MWF 2:20 to 3:20 pm
Class Dates: 21 January through 1 May, 2026
Final Quiz: Wednesday 6 May at 3:00 pm
Text: “AUGSBURGCSC311SteinmetzSpring2026”; zyBooks
Prerequisites: Completion of either CSC 165 or CSC 170 with C- or better

Grading

Grades will be determined based on the accumulation of up to 1000 points. We will have five projects, each worth 30 points (3% of your grade) and a final project worth 100 points for a total of 250 points. There will be thirteen weekly quizzes, each accounting for 30 points. Additionally there will be a final comprehensive quiz worth 140 points. The lowest weekly quiz score will be dropped so quizzes will account for 500 points (360 weekly and 140 final). The remaining 250 points will be given for completing assignments in the zyBooks text and at home along with many class activities.

The entire one thousand points are tallied at the end of the semester. Typically $total \geq 900$ will be various flavors of A grades, $total \geq 800$ B grades, and $total \geq 700$ C grades. $700 > total > 600$ will result in a D grade, while lower will be an F. Earning less than 70% of the quiz points ($quiztotal \leq 350$) will result in a D or F grade.

Please note that a C- or higher is required for a pass under the pass / no-pass grading option and is required for the course to apply to the Computer Science major or minor.

Expectations

- **Attend every class session.** Most of the class sessions will be lecture with class activities while others will be lab sessions. You should bring your laptop during lab sessions, or borrow one from Lindell library.
- **Use class time effectively and allow others to do the same.** Our class time is for discussing, reading, learning, and writing about web applications and databases. It is not time for surfing social networking sites, watching cat videos, or texting friends. No use of phones is allowed in class other than taking photos of the whiteboard. No use of laptops other than writing code or for approved accommodations is permitted.
- **Submit assignments and code when due.** A substantial part of this course is the writing of code of various flavors (html, css, javascript, SQL). In order to remain current, please turn in your code on time. Late assignments will receive no credit. Late projects will receive no credit, but must be submitted anyway. You must be present in class to get credit for in-class activities.
- **Write your own code.** All the work you submit, whether code or other assignments, must be your own work. You are encouraged to discuss ideas, solutions, explanations, and references with your classmates. You are also encouraged to work together as you do assignments and write code, but everything you submit must be your own handiwork. Any time you work together or receive inspiration or help you should cite the source in comments in your code. Please refer to the Augsburg Academic Honesty Policy which can be found at <https://tinyurl.com/AcademicHonestyAugsburg>. To establish a clear division between acceptable and unacceptable sharing, no electronic copying of code is allowed: you must type everything in yourself. Do not send or receive code from your classmates and do not copy-paste code from any other source.
- **Schedule your time.** Reading materials from the text, working on assignments, and writing the substantial amount of code for this course is time consuming. Expect to spend eight hours or more each week reading, writing, and coding outside of class time to succeed in this course.
- **Pay attention to Moodle and email.** Our schedule and assignments are posted on Moodle and may be updated there. I will send email notifying you of any significant changes or updates.

The Online Textbook

The textbook we are using this semester is an online textbook published by zyBooks. You need only a web browser to access this textbook. It includes exercises along with coding exercises and labs which are completed in the book itself. Writing and running code in the online book is separate from the homework and projects we will be doing. The subscription to this text costs \$64 if purchased directly from zyBooks. You may either purchase a subscription key at the bookstore (allowing you to charge this to your account at Augsburg) or directly from zyBooks. Please sign up as soon as you can, and make sure to use your Augsburg email address on your zyBooks account that you create, otherwise you won't be able to access the book. Your subscription will allow access through the end of fall semester 2025.

1. Sign in or create an account at <https://learn.zybooks.com>. You must use your @augsburg.edu address.
2. Enter zyBook code: AUGSBURGCSC311SteinmetzSpring2026
3. Subscribe (with a purchased key or a credit card)

Schedule of Topics

We will be covering all of the visible material in your zyBooks textbook during the course of the semester, along with extra material provided by the instructor.

- Introduction to the web and html
- Introduction to databases and SQL
- Working with html
- Using cascading style sheets (CSS)
- Effects and animation in CSS
- Using Javascript
- The document object model (DOM)
- Using Javascript to manipulate the DOM
- Using Javascript object notation (JSON)
- The AJAX paradigm and web APIs
- Using the jQuery library
- Server side code with Node.js and SQL
- Creating a full-stack application

Additional Information

Accessing Online Resources

Information about getting a computer or an internet connection from the University can be found at <https://sites.augsburg.edu/it/services-we-provide/student-access-to-computing-resources/>.

Disability Accommodations and Accessibility

Students with disabilities who believe that they may need accommodations in this class are encouraged to contact the Center for Learning and Adaptive Student Services (CLASS) (612-330-1053 or stop by the Gage Center welcome desk) as soon as possible to better ensure that such accommodations are implemented in a timely fashion. Contact info about the CLASS program can be found at <http://www.augsburg.edu/class/>. Be aware that it is your responsibility to meet the requirements of your CLASS office accommodations.

Title IX

Augsburg University is committed to creating a safe learning environment for all students, including one that is free of sexual misconduct. If you or someone you know has experienced sexual misconduct, including sexual assault, dating violence, domestic violence, and stalking, you may contact Michael Grewe, Ph.D., LISW, Assistant Provost & Title IX Coordinator, at titleix@augsb.org or 612-330-1499 to report an incident, seek support, and/or take action.

Please be aware that faculty members are required to disclose information about suspected or alleged sexual misconduct or other potential violations of the Augsburg University Sexual Misconduct Policy to the Title IX Coordinator. If the Title IX Coordinator receives information about an incident, they will reach out to offer information about resources, rights, and procedural options as a member of the campus community. If you or another student you know wishes to speak to a confidential resource who does not have this reporting responsibility, you may contact the Center for Wellness and Counseling at cwc@augsb.org or 612-330-1707; Campus Ministry at 612-330-1732; or the Aurora Center at aurora@umn.edu or 612-626-9111 (24/7).

FERPA

As your instructor, I am committed to protecting your privacy by only using university-approved course technologies and adhering to the Family Educational Rights and Privacy Act (FERPA). This includes using your educational data only as allowed by FERPA. I am unable to share any personally identifiable information without your written consent, except for legitimate educational purposes, or under certain conditions described above. FERPA applies to all students enrolled in this institution regardless of sex, race, color, national or ethnic origin, religious belief, age, disability, gender, sexual orientation, gender identity or expression, marital status, familial status, pregnancy, citizenship, creed, genetic information, or veteran status.

Disclaimer

Policies stated within this syllabus are subject to change, following verbal announcement in class. Students are responsible for knowing the current version, always to be found on the course Moodle page.